FEDERAL SPECIFICATION

ISOBUTYL ALCOHOL, (FOR USE IN ORGANIC COATINGS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers the requirements for isobutyl alcohol for use in lacquer thinners.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 141 — Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling, and Testing.
Fed. Test Method Std. No. 791 — Lubricants, Liquid Fuels and Related Products; Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** — The following publications form a part of this specification. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

American Society for Testing Materials Publication:

ASTM Standards—Part 4

(Copies may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

Uniform Freight Classification Ratings, Rules, and Regulations.

(Copies may be obtained from the Uniform Freight Classification Committee, 1 Park Avenue, New York 16, N. Y.)

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3. REQUIREMENTS

3.1 Quantitative requirements. — The isobutyl alcohol shall conform to the requirements specified in table I.

TABLE I.—*Quantitative requirements*

Characteristics	Requirement		Test reference paragraph
	Minimum	Maximum	
Acidity (percent by weight as acetic acid)	----	0.003	4.5.1
Distillation range:			
Initial boiling point °C.	106	----	
Dry point °C.	----	110	4.5.1
Nonvolatile matter (g. per 100 ml.)	----	0.001	4.5.2
Specific gravity at 20°/20°C.	0.802	0.804	4.5.1
Water content (percent)	----	0.2	4.5.4
Color platinum cobalt standard)	----	10	4.5.3

3.2 Qualitative requirements.

3.2.1 *Appearance.* — The isobutyl alcohol shall be clear and free from sediment and suspended matter when examined by transmitted light. (See table II.)

3.2.2 *Odor.* — The isobutyl alcohol shall leave the characteristic odor of isobutyl alcohol and have no residual odor after drying on filter paper for 2 hours. (See table II.)

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 **Sampling.** — Sampling shall be performed in accordance with method 1021 of Federal Test Method Standard No. 141.

4.2.1 *For Military agencies.*—Sampling for Military agencies shall be performed in accordance with method 1031 of Federal Test Method Standard No. 141.

4.3 **Inspection.** — Inspection shall be performed in accordance with method 1021 of Federal Test Method Standard No. 141.

4.3.1 *For Military agencies.* — Inspection shall be performed in accordance with method 1031 of Federal Test Method Standard No. 141.

4.4 **Resubmitted inspection lots.** — Paragraph 12.2 of Military Standard MIL–STD–105 shall apply except that a resubmitted inspection lot shall be inspected by the contractor under supervision of the Government Inspector using tightened inspection. For visual examination, where the original acceptance number was zero, a sample size represented by the next higher sample size code letter shall be chosen. Before an inspection lot is resubmitted, full particulars concerning the cause of previous rejection and the action taken to correct the defects found in the inspection lot shall be furnished by the contractor to the Government inspector.

4.5 **Tests.** — Tests shall be made in accordance with the applicable methods of Federal Test Method Standard No. 141, and as specified herein.

4.5.1 The tests as listed in table II shall be conducted in accordance with Federal Test Method Standard No. 141.

4.5.2 *Nonvolatile matter.*—Place 100 milliliters of the sample in a tared evaporating dish and evaporate almost to dryness over a steam bath. Heat the dish and residue in an oven at 105° ± 2°C. (221° ± 4°F.) to con-

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TABLE II.

Tests	Method No.
Acidity (See 3.1)	5252
Specific gravity (any convenient method enumerated) (See 3.1)	4138
Appearance (See 3.2.1)	4261
Odor (See 3.2.2)	4401
Distillation range (See 3.1)	4301

stant weight, cool in a desiccator and weigh. Report the difference in weight as grams of nonvolatile matter per 100 milliliter of the sample. (See 3.1.)

4.5.3 *Color.*—The color of the isobutyl alcohol shall be determined in accordance with method D 1209–54 of the American Society for Testing Materials Standards. (See 3.1.)

4.5.4 *Water content.*—Water content of the isobutyl alcohol shall be determined in accordance with method 3253 of Federal Test Method Standard No. 791. (See 3.1.)

5. PREPARATION FOR DELIVERY

For civil agency procurements, the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Level C.*—Isobutyl alcohol in quantity specified shall be packaged in accordance with the manufacturers' commercial practice.

5.2 Packing.

5.2.1 *Level C.*—Isobutyl alcohol packaged as specified shall be packed in containers which will insure acceptance by common carrier and safe delivery to destination at the lowest applicable rate. Shipping containers shall conform to the Uniform Freight Classification Rating Rules and Regulations or other regulations as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.*—Marking for civil agencies shall be in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*—In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use. — This specification covers isobutyl alcohol, an ingredient for use in specifications for paints and related materials and as such is not intended for use for direct procurement by the Government. In the event of small purchases for experimental purposes and for purchases by industry, suitable preparation for delivery requirements should be made a part of the contract. The quality assurance provisions are provided for in connection with inspection of material when used as an ingredient in paints purchased for the Government.

6.2 Basis of purchase. — The material shall be purchased by volume, the unit being a U. S. gallon (231 cubic inches at 15.5°C. (60°F.)).

6.3 Ordering data. — Procurement documents should state the title, number, and date of this specification.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
Navy—A Sh.

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